

Bingyixuan(Mason) Dong

bingyxdong@gmail.com | bingyixuan.com | 9099055111

EDUCATION

Cornell University

Master of Engineering in Computer Science

Ithaca, NY

Expected May. 2027

University of Wisconsin - Madison

Bachelor of Science in Computer Sciences (Honors) and Data Science.

Madison, WI

May. 2026

Minors in Educational Policy Studies, Education and Educational Services.

CS Honors thesis: *Enhancing Visualization in Computer Science Education With AI-Powered Visual Generation*

Advisor: Prof. Bilge Mutlu

Relevant Coursework: Artificial Intelligence, User Interfaces, Virtual Reality, Algorithms, Monte Carlo Simulation for Quantitative Research, Teaching & Learning in Diverse Settings.

RESEARCH INTERESTS

My main research interests are to explore the computational foundations of reliable AI powered education technology that are safe, personalized, and cognitively engaging. Researching in the interdisciplinary field of Artificial Intelligence and Education. Mainly targeting personalized Artificial Intelligence, framework to reduce hallucinations, and engaging education technology.

RESEARCH EXPERIENCE

Independent Honors Thesis Research [\[Poster\]](#)

Madison, WI

Undergraduate Research Assistant **PI: Prof. Bilge Mutlu**

Sep. 2025 – Present

- Conducted 10 IRB-approved semi-structured interviews with CS instructors and TAs, using thematic analysis to identify visualization challenges in teaching abstract CS concepts.
- Developed a web interface that uses AI image generation to help educators create custom visualizations of algorithms, data structures, and system architecture, and ran user testing with CS educators to evaluate usability and effectiveness.

Madison Computer Sciences Education (MadCSE) Lab

Madison, WI

Undergraduate Research Assistant **PI: Prof. Meenakshi Syamkumar**

Literature Review on tools in CS Education

Sep. 2025 – Present

- Conducting systematic analysis for 4,500+ research papers to identify emerging LLM applications and pedagogical approaches in CS.
- Analyzing papers across NLP, HCI, and learning sciences to identify prevalent approaches and emerging techniques for LLM integration in programming education.
- Synthesizing findings into a comprehensive taxonomy; targeting 2026 submission to SIGCSE.

LLM Course Analysis Agent

Sep. 2025 – Present

- Built Python pipeline integrating GPT-4 API to process lecture transcripts and homework assignments for learning objective coverage analysis.
- Architected an end-to-end RAG pipeline in Python to audit curriculum integrity.
- Designed prompt engineering workflow to automatically identify over-covered, under-covered, and missing learning objectives across course materials.

People and Robots Laboratory

Madison, WI

Undergraduate Research Assistant **PI: Prof. Bilge Mutlu**

Learning Exploration Robot

July. 2025 – Aug. 2025

- Supported field study implementation across different families, managing robot setup and documenting child-robot interactions.

- Performed qualitative analysis on conversational transcripts to identify learning topics and interaction patterns in child-robot dialogues.
- Identified key interaction patterns and linguistic cues that correlate with sustained student engagement and "curiosity-driven" learning.

Robot Plan B

May. 2025 – Aug. 2025

- Engineered a full-stack web interface using React and JavaScript that enables non-expert operators to author complex robot "fallback" sequences via a drag-and-drop block-programming paradigm.
- Developed a robust communication layer between the React frontend and a Flask-based backend, facilitating real-time execution of robot manipulation and navigation commands.

Mathematical Action and Gesture in Instruction and Cognition (MAGIC) Lab

Madison, WI

Undergraduate Research Assistant **PI: Prof. Mitchell J. Nathan**

Immersive Geometric Reasoning

Oct. 2025 – Present

- Designing VR activities and assessment tools to measure learning outcomes and engagement.
- Running study sessions to investigate how physical body movements affect geometric reasoning and mathematical cognition in VR-based learning environments.

ICE Lab

Madison, WI

Undergraduate Research Assistant **PI: Prof. Icy Zhang**

Investigating Student Questioning and Engagement Patterns in LLM Tutoring

Sep. 2025 – Present

- Designed and implemented the front-end interface for a mixed-methods study contrasting LLM and human instruction.
- Identify pedagogical differences in student question patterns and motivation between LLM and human instruction.

TEACHING EXPERIENCE

University of Wisconsin-Madison Department of Computer Sciences

Madison, WI

Peer Mentor, (CS639 - Data Management for Data Science)

Spring 2024, Spring 2025

- Conducted biweekly office hours, providing direct assistance to students in course on database systems, including MySQL, MongoDB, and Elasticsearch.
- Guide students in exploratory data analysis, machine learning techniques such as ARIMA, LightGBM, And XGBoost.
- Designing and deploying novel course projects focused on real-world data science workflows.

PROFESSIONAL EXPERIENCE

CrissCross Express

Los Angeles, CA

Software Engineering Intern (Summers 2023 & 2024)

May. 2023 – Aug. 2024

- Developed package tracking system, integrating barcode scanning with SQL database, enabling real-time shipment monitoring and location tracking between Mainland China and U.S.
- Engineered backend scripts linking RESTful APIs with SQL databases to automate shipment updates and ensure reliable synchronization across all warehouses.

University of Wisconsin-Madison University Housing

Madison, WI

Advisory Board Member

Oct. 2023 – May. 2024

- Selected to represent DeJope Hall's 800 + residents on university housing advisory board.
- Advocating for student needs and collaborating with administrators to enhance residential learning and living experiences.

HONORS & AWARDS

University of Wisconsin-Madison Dean's List Award

2024

President Silver Volunteer Service Award

2022

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